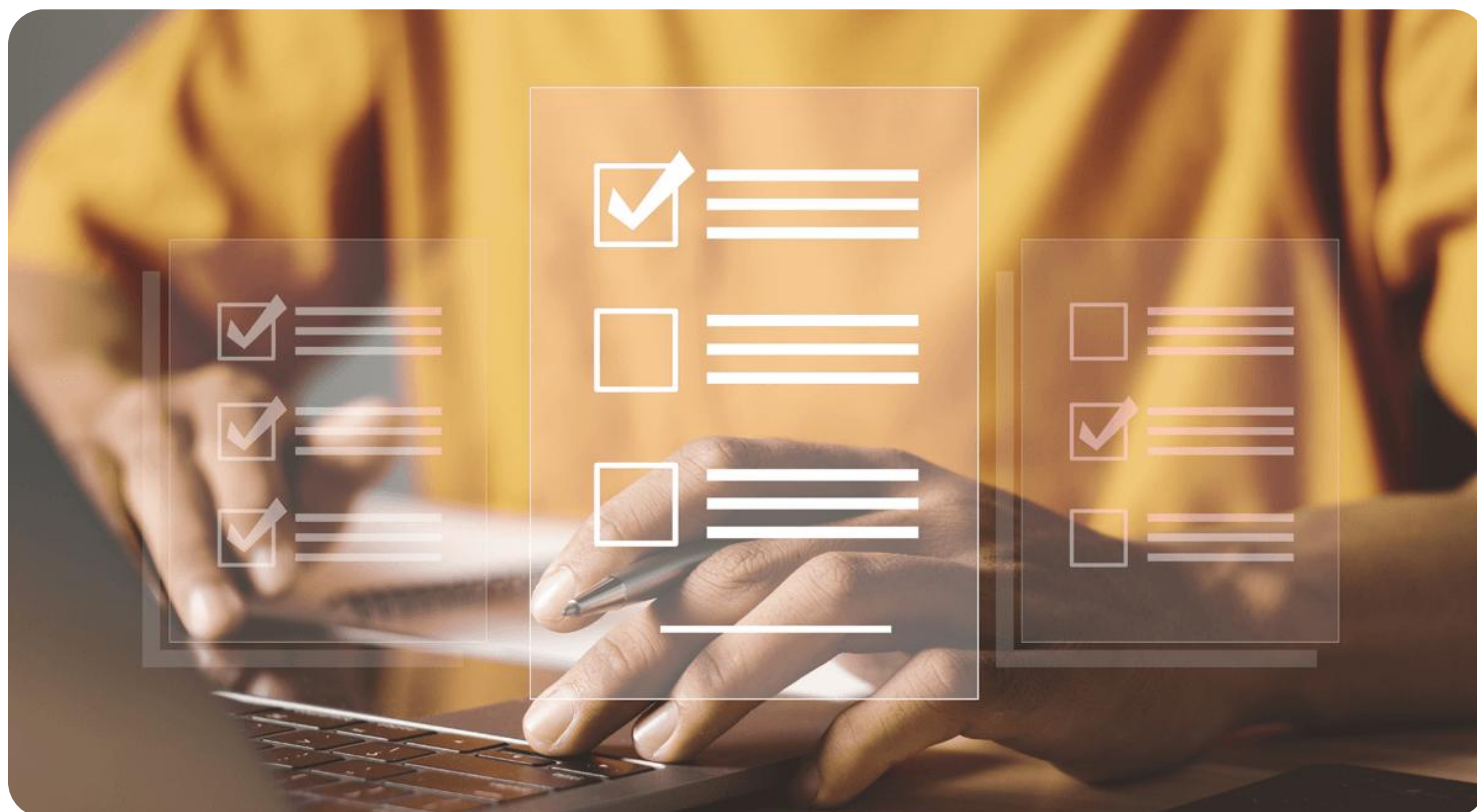




Semana 5

Desarrollo Backend III (PBY2203)

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Evidencia 1

```
PS C:\Users\pedro\OneDrive\Escritorio\Duoc Bimestre 8\Backend-3\Semana_5\Exp2_S5_Pedro_Breit\bff-web> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
09a2ae5faf42   postgres:16   "docker-entrypoint.s..." About an hour ago Up About an hour (healthy) 0.0.0.0:5433->5432/tcp, [::]:5433->5432/tcp banco-xyz-postgres
PS C:\Users\pedro\OneDrive\Escritorio\Duoc Bimestre 8\Backend-3\Semana_5\Exp2_S5_Pedro_Breit\bff-web>
```

Ejecución del contenedor PostgreSQL 16 mediante Docker, utilizado como base de datos local del Backend Central en el puerto 5433.

Evidencia 2

```
2026-09-08T17:09:35.430-03:00 INFO 30132 --- [banco-central-xyz] [main] c.d.b.BancoCentralXyzApplication : Starting BancoCentralXyzApplication using Java 21.0.10 with PID 30132 (C:\Users\pedro\OneDrive\Escritorio\Duoc Bimestre 8\Backend-3\Semana_5\Exp2_S5_Pedro_Breit\Banco-central\banco-central-xyz)
2026-09-08T17:09:35.433-03:00 INFO 30132 --- [banco-central-xyz] [main] c.d.b.BancoCentralXyzApplication : No active profile set, falling back to 1 default profile: "default"
2026-09-08T17:09:35.903-03:00 INFO 30132 --- [banco-central-xyz] [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-09-08T17:09:35.951-03:00 INFO 30132 --- [banco-central-xyz] [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 42 ms. Found 4 JPA repository interfaces
2026-09-08T17:09:36.344-03:00 INFO 30132 --- [banco-central-xyz] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-09-08T17:09:36.354-03:00 INFO 30132 --- [banco-central-xyz] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-09-08T17:09:36.355-03:00 INFO 30132 --- [banco-central-xyz] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.24]
2026-09-08T17:09:36.400-03:00 INFO 30132 --- [banco-central-xyz] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 931 ms
2026-09-08T17:09:36.406-03:00 INFO 30132 --- [banco-central-xyz] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-09-08T17:09:36.597-03:00 INFO 30132 --- [banco-central-xyz] [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection org.postgresql.jdbc.PgConnection@517704
2026-09-08T17:09:36.598-03:00 INFO 30132 --- [banco-central-xyz] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-09-08T17:09:36.692-03:00 INFO 30132 --- [banco-central-xyz] [main] org.hibernate.orm.jpa : HHH000540: Processing PersistenceUnitInfo [name: default]
2026-09-08T17:09:36.736-03:00 INFO 30132 --- [banco-central-xyz] [main] org.hibernate.orm.core : HHH000001: Hibernate ORM core version 7.4.5.Final
2026-09-08T17:09:37.169-03:00 INFO 30132 --- [banco-central-xyz] [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-09-08T17:09:37.245-03:00 INFO 30132 --- [banco-central-xyz] [main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [jdbc:postgresql://localhost:5433/banco_xyz]
Database driver: PostgreSQL JDBC Driver
Database dialect: PostgreSQLDialect
Database version: 16.15
Default catalog/schema: banco_xyz/public
Autocommit mode: undefined/unknown
Isolation level: READ_COMMITTED [default READ_COMMITTED]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-09-08T17:09:38.000-03:00 INFO 30132 --- [banco-central-xyz] [main] org.hibernate.orm.core : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable)
Hibernate:
alter table if exists usuarios
alter column rol set data type varchar(255)
Hibernate:
alter table if exists usuarios
alter column username set data type varchar(255)
2026-09-08T17:09:38.137-03:00 INFO 30132 --- [banco-central-xyz] [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-09-08T17:09:38.196-03:00 INFO 30132 --- [banco-central-xyz] [main] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will be used.
2026-09-08T17:09:38.416-03:00 WARN 30132 --- [banco-central-xyz] [main] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering.
n-view to disable this warning
2026-09-08T17:09:38.495-03:00 INFO 30132 --- [banco-central-xyz] [main] r$InitializeUserDetailsManagerConfigurer : Global AuthenticationManager configured with UserDetailsService bean with name custom
2026-09-08T17:09:38.838-03:00 INFO 30132 --- [banco-central-xyz] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path '/'
2026-09-08T17:09:38.842-03:00 INFO 30132 --- [banco-central-xyz] [main] c.d.b.BancoCentralXyzApplication : Started BancoCentralXyzApplication in 3.707 seconds (process running for 3.993)
```

Inicio exitoso del **Backend Central**, ejecutándose en el puerto 8081 y conectado correctamente a la base de datos PostgreSQL.

Evidencia 3

```
2026-09-08T17:12:24.896-03:00 INFO 28636 --- [bff-cajeros] [main] c.d.bff-cajeros.BffCajerosApplication : Starting BffCajerosApplication using Java 21.0.10 with PID 28636 (C:\Users\pedro\OneDrive\Escritorio\Duoc Bimestre 8\Backend-3\Semana_5\Exp2_S5_Pedro_Breit\bff-cajeros)
2026-09-08T17:12:24.899-03:00 INFO 28636 --- [bff-cajeros] [main] c.d.bff-cajeros.BffCajerosApplication : No active profile set, falling back to 1 default profile: "default"
2026-09-08T17:12:25.638-03:00 INFO 28636 --- [bff-cajeros] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8082 (https)
2026-09-08T17:12:25.650-03:00 INFO 28636 --- [bff-cajeros] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-09-08T17:12:25.651-03:00 INFO 28636 --- [bff-cajeros] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.24]
2026-09-08T17:12:25.687-03:00 INFO 28636 --- [bff-cajeros] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 752 ms
2026-09-08T17:12:25.978-03:00 WARN 28636 --- [bff-cajeros] [main] .s.a.UserDetailsServiceAutoConfiguration :
Using generated security password: 8035aff5-1fe2-4df0-a113-e24668ef4699
This generated password is for development use only. Your security configuration must be updated before running your application in production.
2026-09-08T17:12:25.983-03:00 INFO 28636 --- [bff-cajeros] [main] r$InitializeUserDetailsManagerConfigurer : Global AuthenticationManager configured with UserDetailsService bean with name inMemoryUserDetailsService
2026-09-08T17:12:26.296-03:00 INFO 28636 --- [bff-cajeros] [main] o.a.t.util.net.NioEndpoint.certificate : Connector [https-jsse-nio-8082], TLS virtual host [_default_], certificate type [UNDEFINED]
ias [bff-cajeros] with trust store [null]
2026-09-08T17:12:26.302-03:00 INFO 28636 --- [bff-cajeros] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8082 (https) with context path '/'
2026-09-08T17:12:26.306-03:00 INFO 28636 --- [bff-cajeros] [main] c.d.bff-cajeros.BffCajerosApplication : Started BffCajerosApplication in 1.712 seconds (process running for 1.968)
```

Inicio exitoso del **BFF Cajero**, ejecutándose mediante HTTPS en el puerto 8082 con configuración de certificado SSL local.

Evidencia 4

```

2026-09-08T17:14:53.614-03:00 INFO 4490 --- [bff-mobile] [ main] c.duoc.bff.mobile.BffMobileApplication : Starting BffMobileApplication using Java 21.0.10 with PID 4400 (C:\Users\pedro\OneDrive\Escritorio\Duoc Bimestre 8\Backend-3\Semana 5\Exp2_S5_Pedro_Breit\bff-mobile)
2026-09-08T17:14:53.616-03:00 INFO 4490 --- [bff-mobile] [ main] c.duoc.bff.mobile.BffMobileApplication : No active profile set, falling back to 1 default profile: "default"
2026-09-08T17:14:54.353-03:00 INFO 4490 --- [bff-mobile] [ main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (https)
2026-09-08T17:14:54.364-03:00 INFO 4490 --- [bff-mobile] [ main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-09-08T17:14:54.365-03:00 INFO 4490 --- [bff-mobile] [ main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.24]
2026-09-08T17:14:54.406-03:00 INFO 4490 --- [bff-mobile] [ main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 752 ms
2026-09-08T17:14:54.682-03:00 WARN 4490 --- [bff-mobile] [ main] s.a.UserDetailsServiceAutoConfiguration :

Using generated security password: 0c893084-acde-4763-8197-42401ff8a8b5

This generated password is for development use only. Your security configuration must be updated before running your application in production.

2026-09-08T17:14:54.688-03:00 INFO 4490 --- [bff-mobile] [ main] r.s.InitializeUserDetailsServiceConfigurer : Global AuthenticationManager configured with UserDetailsServiceImpl bean with n
2026-09-08T17:14:55.005-03:00 INFO 4490 --- [bff-mobile] [ main] o.a.t.util.net.NioEndpoint.certificate : Connector [https-jsse-nio-8083], TLS virtual host [_default], certificate
s [bff-mobile] with trust store [null]
2026-09-08T17:14:55.013-03:00 INFO 4490 --- [bff-mobile] [ main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (https) with context path '/'
2026-09-08T17:14:55.016-03:00 INFO 4490 --- [bff-mobile] [ main] c.duoc.bff.mobile.BffMobileApplication : Started BffMobileApplication in 1.701 seconds (process running for 1.96)

```

Inicio exitoso del **BFF Mobile**, ejecutándose mediante HTTPS en el puerto 8083 con configuración de certificado SSL local.

Evidencia 5

```

2026-09-08T17:16:58.253-03:00 INFO 30468 --- [bfff-web] [main] com.duoc.bfff_web.BfffWebApplication : Starting BfffWebApplication using Java 21.0.10 with PID 30468 (C:\Users\pedro\OneDrive\Escritorio\Duoc Bimestre 8\Back-end-3\Semana 5\Exp2_S5_Pedro_Breit\bfff-web)
getClasses started by pedro in C:\Users\pedro\OneDrive\Escritorio\Duoc Bimestre 8\Back-end-3\Semana 5\Exp2_S5_Pedro_Breit\bfff-web)
2026-09-08T17:16:58.255-03:00 INFO 30468 --- [bfff-web] [main] com.duoc.bfff_web.BfffWebApplication : No active profile set, falling back to 1 default profile: "default"
2026-09-08T17:16:58.989-03:00 INFO 30468 --- [bfff-web] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8084 (https)
2026-09-08T17:16:59.002-03:00 INFO 30468 --- [bfff-web] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-09-08T17:16:59.002-03:00 INFO 30468 --- [bfff-web] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.24]
2026-09-08T17:16:59.044-03:00 INFO 30468 --- [bfff-web] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 749 ms
2026-09-08T17:16:59.308-03:00 WARN 30468 --- [bfff-web] [main] .s.a.UserDetailsServiceAutoConfiguration :

Using generated security password: 5f21073e-0d00-49cc-b9e1-b0aec95e4160

This generated password is for development use only. Your security configuration must be updated before running your application in production.

2026-09-08T17:16:59.313-03:00 INFO 30468 --- [bfff-web] [main] r$IInitializeUserDetailsServiceConfigurer : Global AuthenticationManager configured with UserDetailsService bean with name
2026-09-08T17:16:59.654-03:00 INFO 30468 --- [bfff-web] [main] o.a.t.util.net.NioEndpoint.certificate : Connector [https-jssse-nio-8084], TLS virtual host [_default_], certificate type
[bfff-web] with trust store [null]
2026-09-08T17:16:59.665-03:00 INFO 30468 --- [bfff-web] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8084 (https) with context path '/'
2026-09-08T17:16:59.670-03:00 INFO 30468 --- [bfff-web] [main] com.duoc.bfff_web.BfffWebApplication : Started BfffWebApplication in 1.734 seconds (process running for 2.055)

```

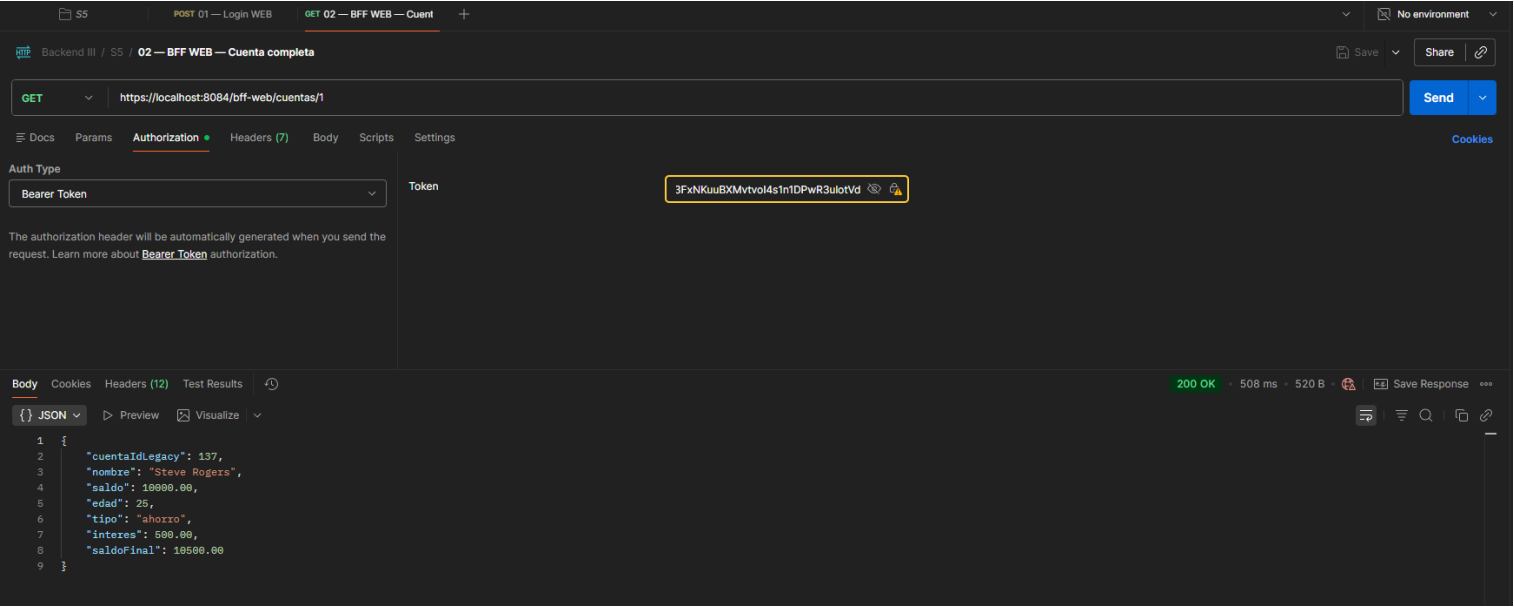
Inicio exitoso del **BFF Web**, ejecutándose mediante HTTPS en el puerto 8084 con configuración de certificado SSL local.

Evidencia 6



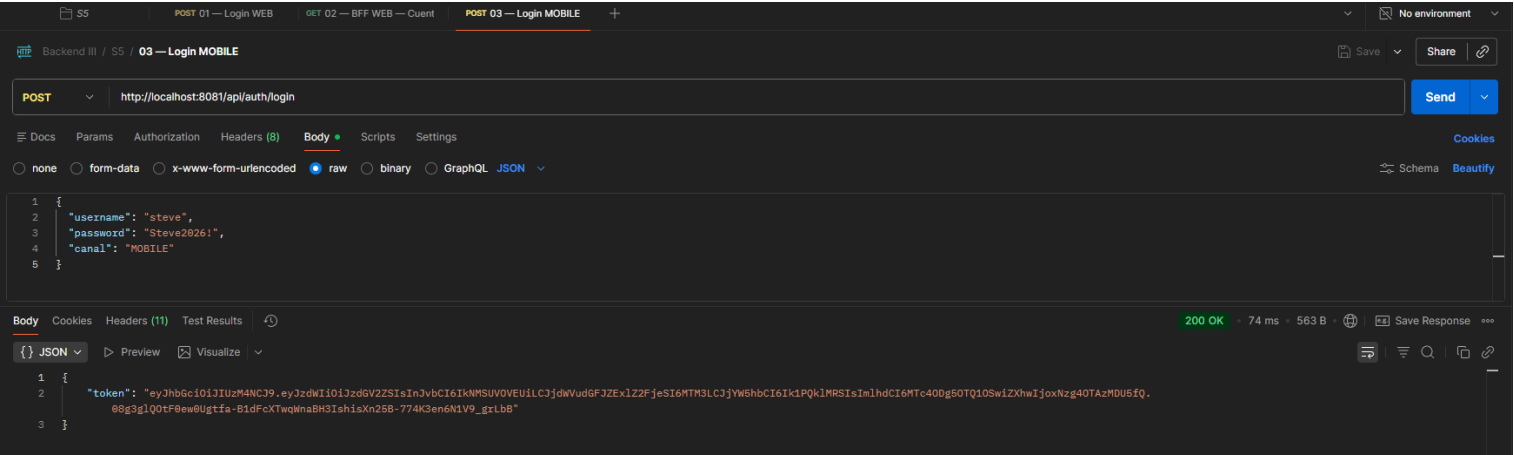
Autenticación exitosa desde el canal **Web**, obteniendo un token JWT válido mediante el endpoint de login del Backend Central.

Evidencia 7



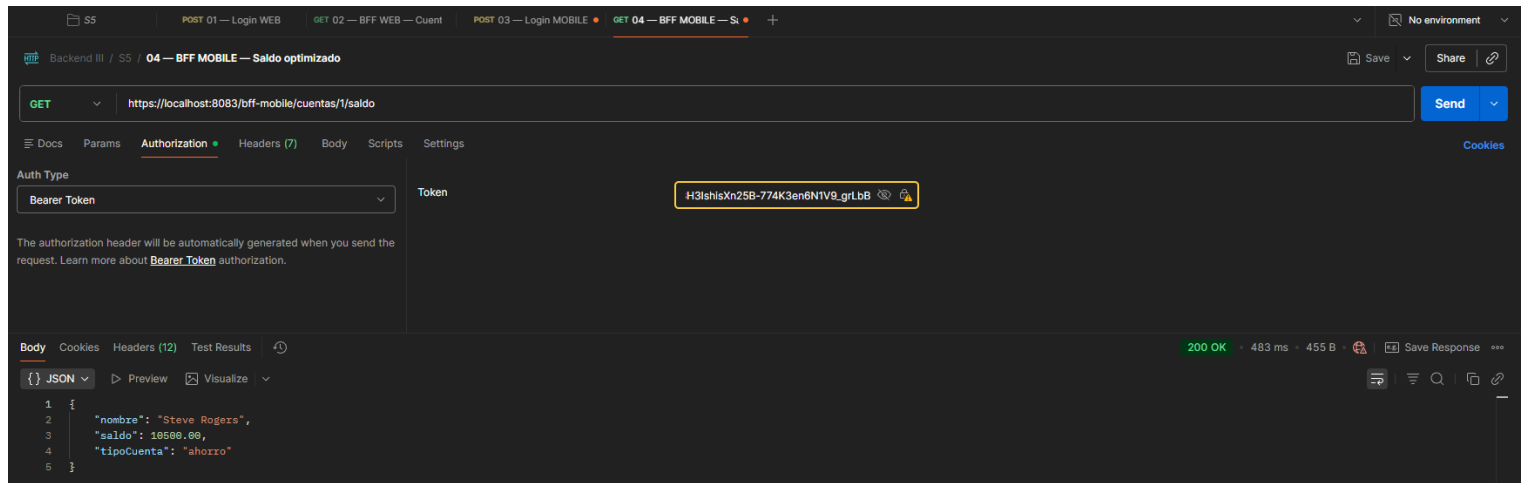
Consulta exitosa desde el **BFF Web**, mostrando la información completa de la cuenta bancaria y confirmando la adaptación de la respuesta para este canal.

Evidencia 8



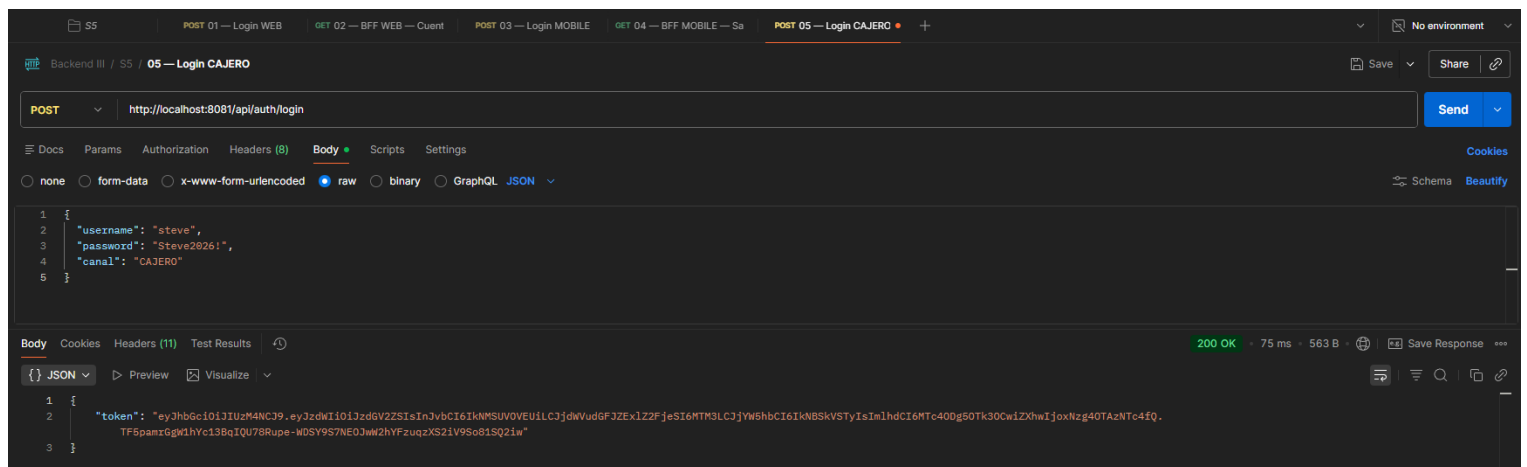
Autenticación exitosa desde el canal **Mobile**, obteniendo un token JWT válido asociado específicamente a este canal.

Evidencia 9



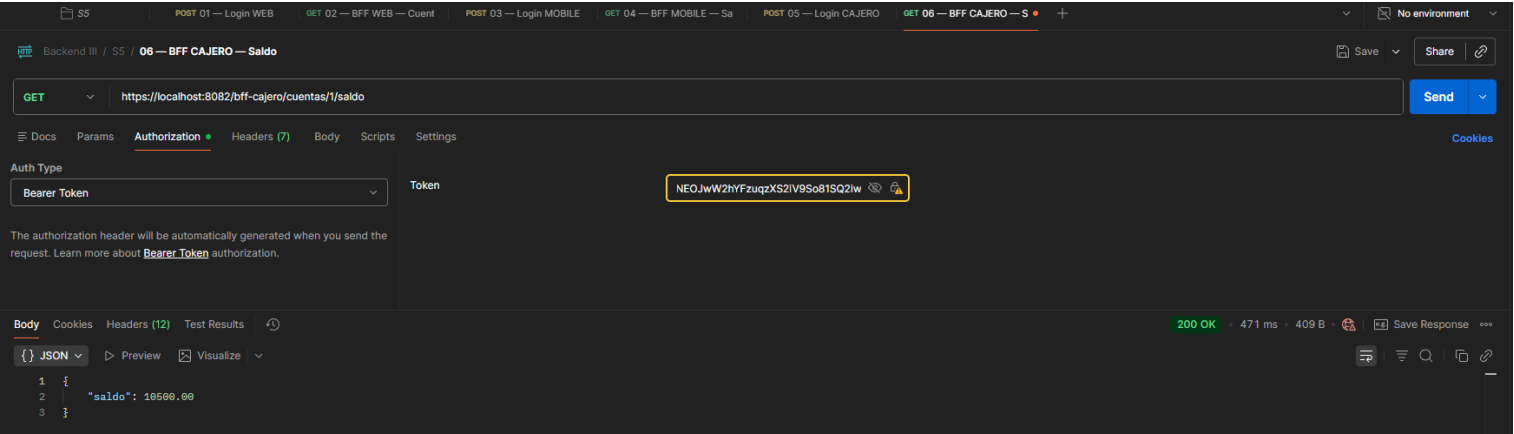
Consulta exitosa desde el **BFF Mobile**, mostrando una respuesta optimizada con únicamente la información esencial de la cuenta para el canal móvil.

Evidencia 10



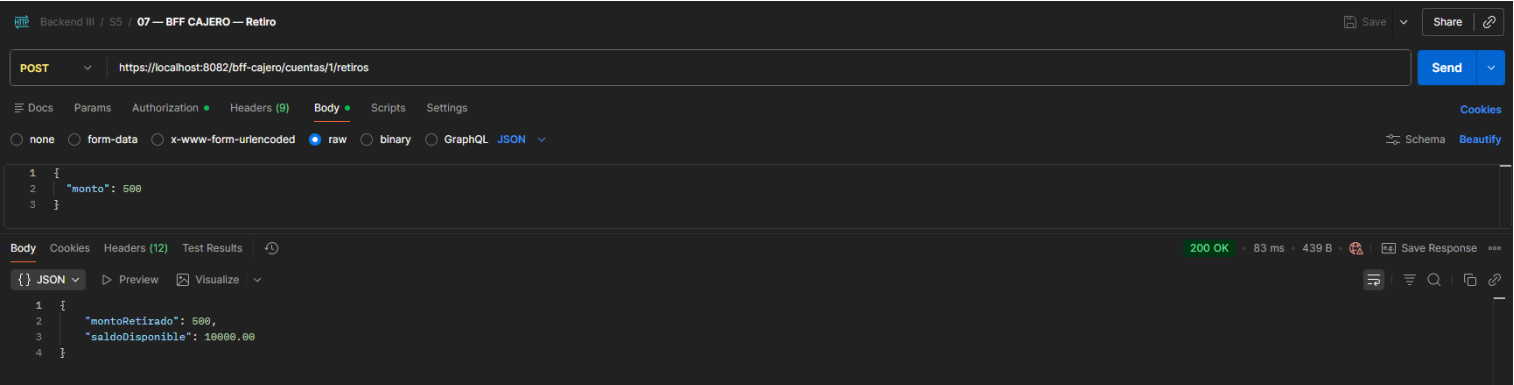
Autenticación exitosa desde el canal **Cajero**, obteniendo un token JWT válido asociado específicamente al BFF de cajeros.

Evidencia 11



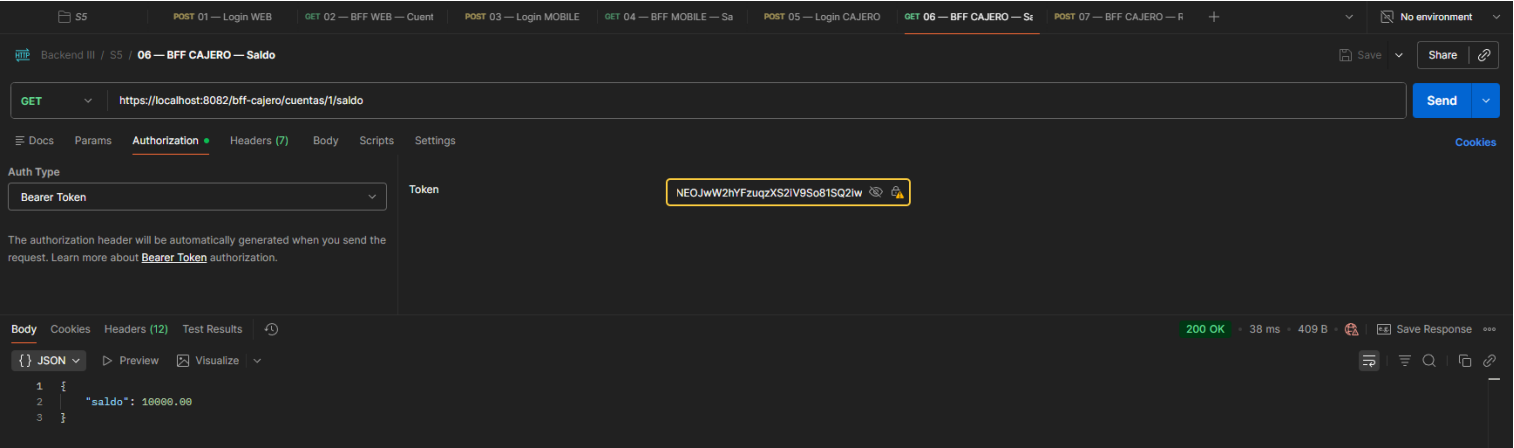
Consulta exitosa del saldo desde el **BFF Cajero**, mostrando únicamente la información necesaria para este canal y validando el acceso mediante un token JWT de tipo CAJERO.

Evidencia 12



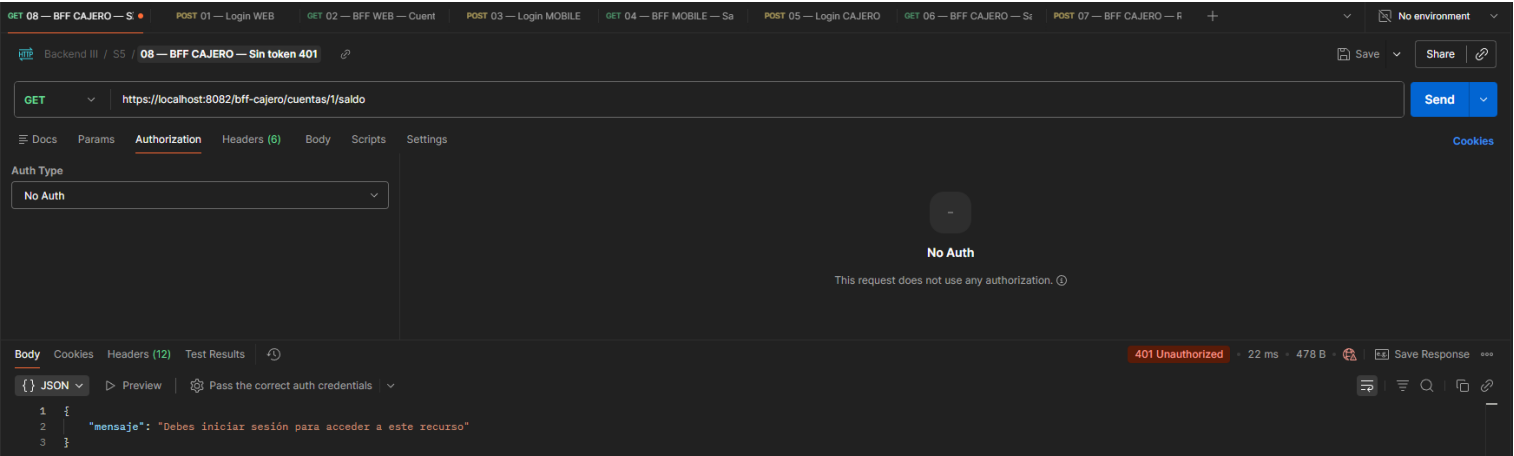
Retiro realizado exitosamente desde el **BFF Cajero**, procesando una operación de \$500 y retornando el monto retirado junto con el saldo disponible actualizado.

Evidencia 13



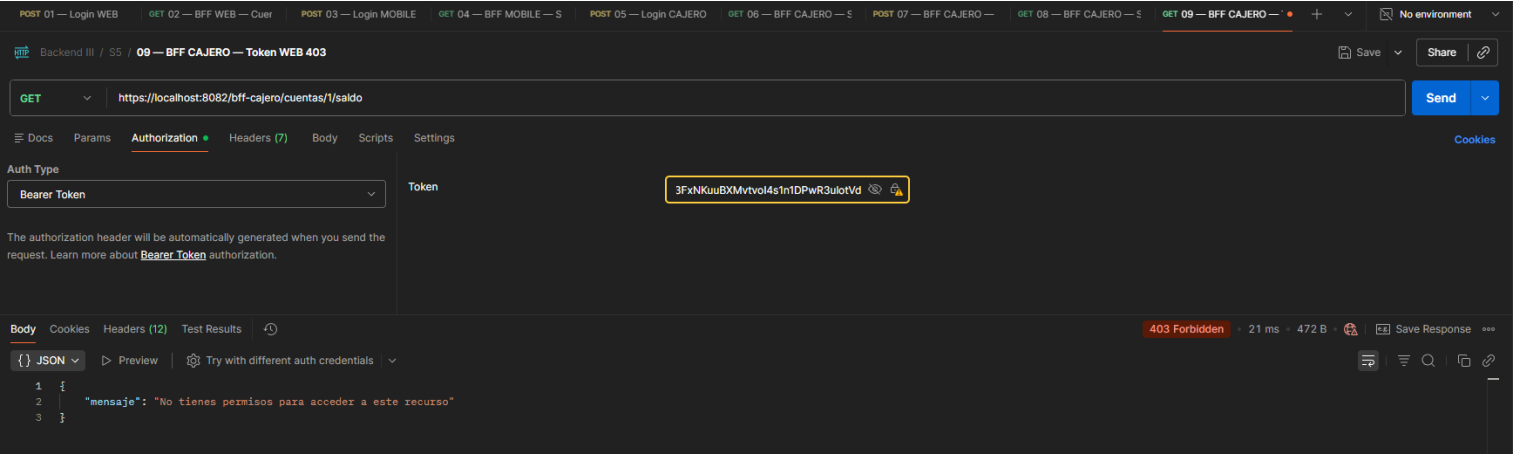
Consulta del saldo posterior al retiro desde el **BFF Cajero**, confirmando que la operación fue persistida correctamente y que el nuevo saldo corresponde a \$10.000.

Evidencia 14



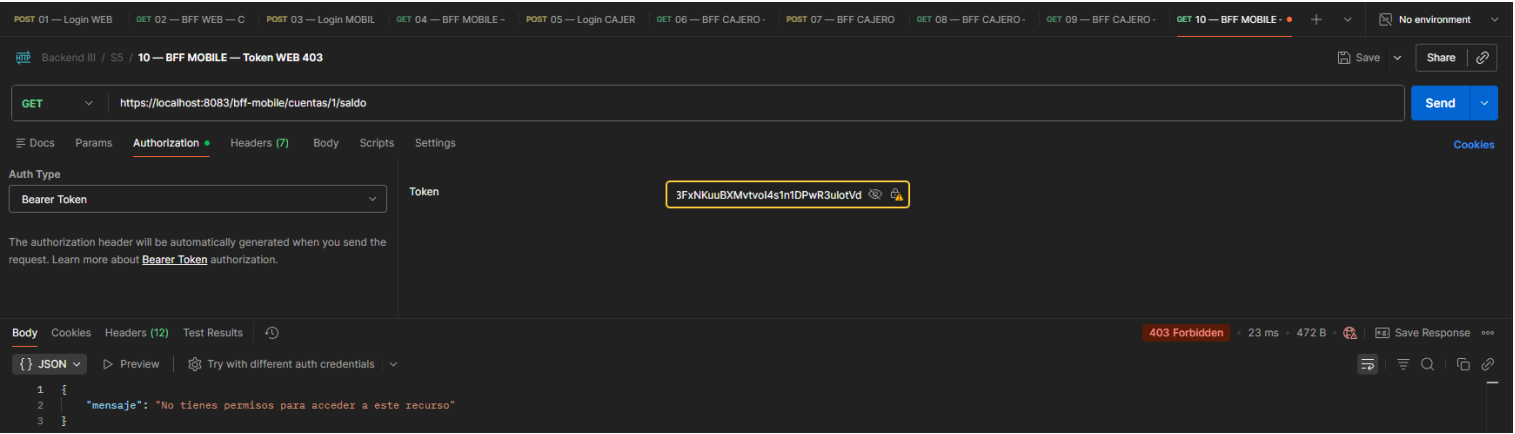
Intento de acceso al **BFF Cajero sin token JWT**, obteniendo una respuesta 401 Unauthorized y validando la protección del endpoint.

Evidencia 15



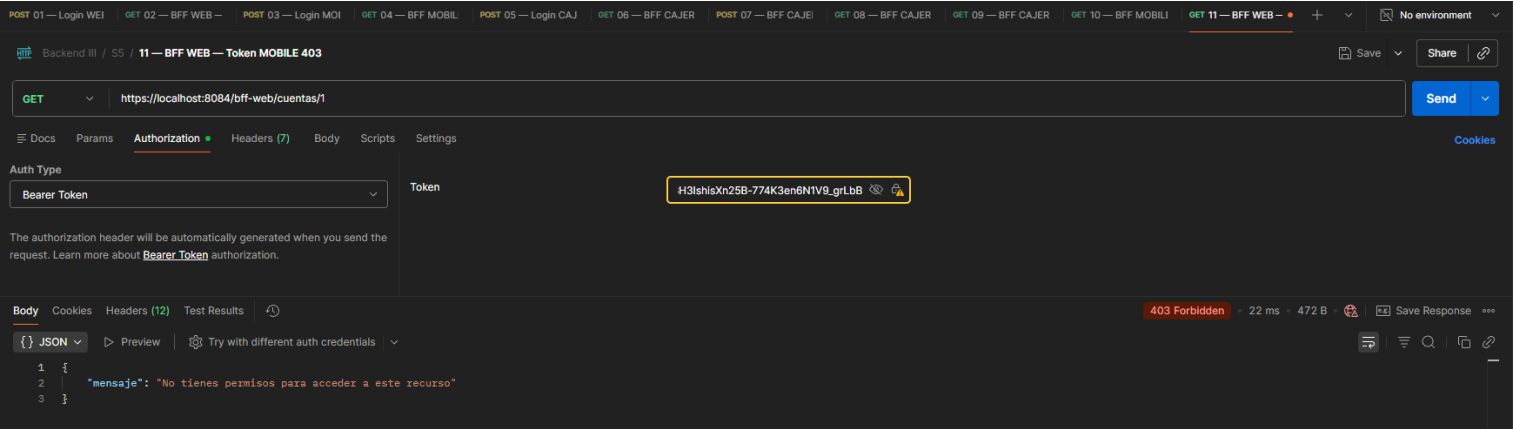
Intento de acceso al **BFF Cajero** utilizando un token del canal Web, obteniendo 403 Forbidden y demostrando la autorización específica por canal.

Evidencia 16



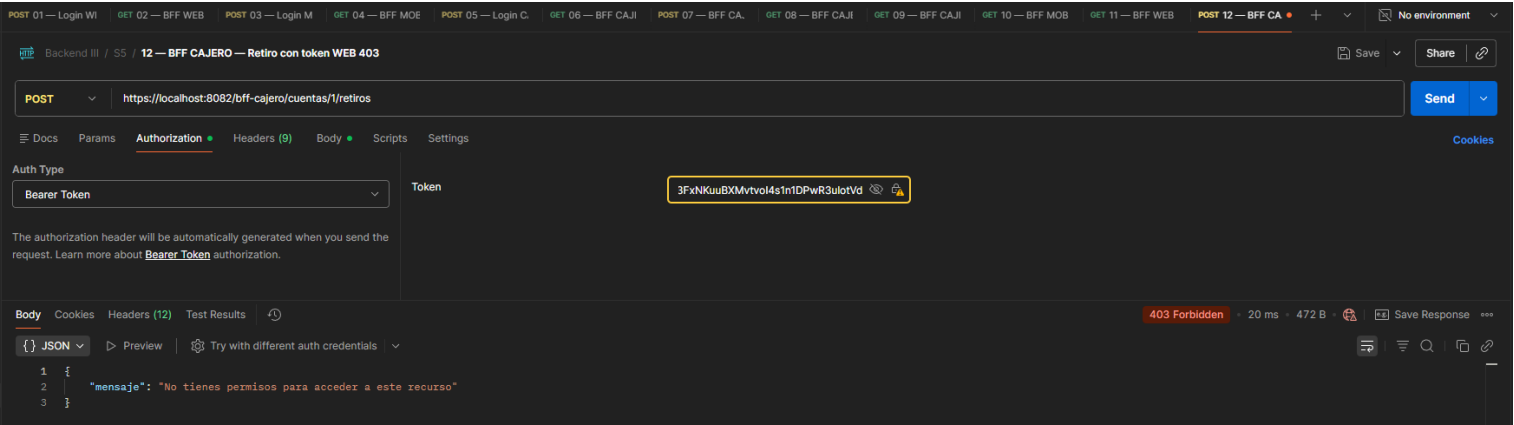
Intento de acceso al **BFF Mobile** utilizando un token del canal Web, obteniendo 403 Forbidden y confirmando la restricción de acceso entre canales.

Evidencia 17



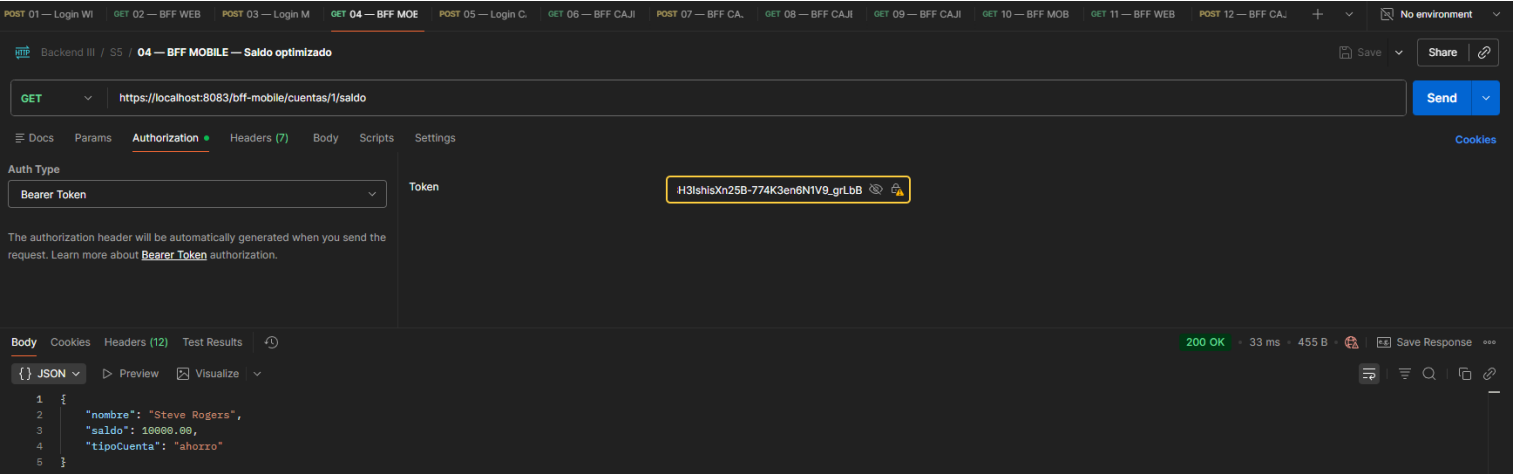
Intento de acceso al **BFF Web** utilizando un **token del canal Mobile**, obteniendo 403 Forbidden y comprobando que cada BFF acepta únicamente tokens correspondientes a su canal.

Evidencia 18



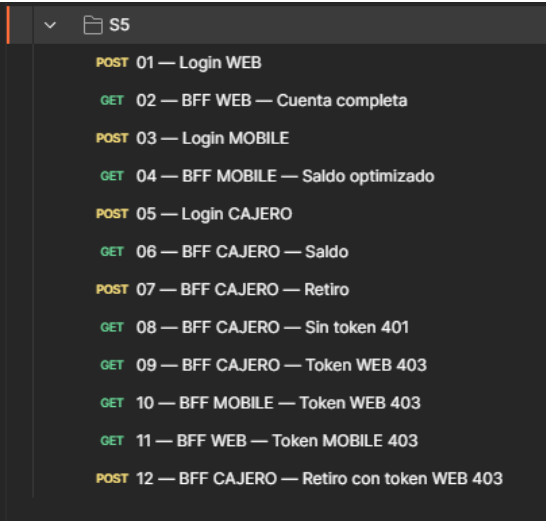
Intento de realizar un retiro en el **BFF Cajero** utilizando un **token del canal Web**, obteniendo 403 Forbidden y evitando la ejecución de una operación crítica desde un canal no autorizado.

Evidencia 19



Consulta desde el **BFF Mobile posterior al retiro**, mostrando el saldo actualizado y comprobando que los distintos BFF consumen la misma información persistida por el Backend Central.

Evidencia 20



Vista general de la **colección de pruebas en Postman**, donde se agrupan las solicitudes utilizadas para validar autenticación, consultas, retiro, seguridad y restricciones de acceso de los tres BFF.